How do women and men perform in each subject, and could it shape who chooses which major?

Subject-wise GPA by gender in the first four semesters, six batches

1. What the data are

Our Dean of Academic Affairs (Dr. Rangeet Bhattacharyya) gave us the mean GPA of men and of women in seven subject areas (physics, chemistry, biology, earth sciences, mathematics, computer science and humanities) for six batches. Batches are labelled Batch 3 to Batch 8, matching the labels used in our CGPA report; the original codes are kept out of this report.

The numbers cover only the **first four semesters**. In semesters 1 and 2 every student studies every subject. In semesters 3 and 4 each student studies three subjects of her or his choice. Majors are chosen after semester 4. So each figure is an average over the courses students took in that subject, in the two semesters when everyone took it plus any later semester when they chose it. This is the period in which students form an impression of where they are strong, just before choosing a major, which is why it is useful here.

We have one number per gender, subject and batch (84 numbers in all). We do not have the number of students or the spread of marks, so we can compare averages across batches but cannot test differences within a batch.

Note: "Gap" below always means **women minus men**, in GPA points, so a positive gap means women's average was higher.

2. What we found

- **Overall, women and men do about equally well.** Averaged over all seven subjects and six batches, the gap is +0.04 points. It was positive in Batches 3 and 4 (+0.28 and +0.41) and slightly negative in Batches 5 to 8 (−0.22 to −0.05).
- **Humanities is the one clear difference.** Women's average was higher in all six batches (average gap +0.16, 95% interval +0.06 to +0.26). No other subject shows a consistent direction.
- **Biology and earth sciences lean towards women** (+0.25 and +0.24 on average; women ahead in 4 of 6 batches each), **physics, mathematics and computer science lean towards men** (-0.18, -0.15 and -0.12; women ahead in only 2, 2 and 1 of 6 batches). In each of these the interval includes zero, so with six batches we cannot rule out chance for any single subject.

- **Taken together, the pattern is consistent.** Women's average gap in biology, earth sciences and humanities was higher than in physics and mathematics in 5 of 6 batches, by 0.38 points on average ($p = 0.02$, paired t-test over batches; $p = 0.06$, rank test, the smallest p possible with six batches). This is suggestive rather than firm.
- **Mathematics is the lowest-scoring subject for both genders** (average rank 6.7 for men and 6.8 for women, of 7). Both men and women do worst in it, so women's lower relative performance there is not a case of women failing where men succeed. However, the data for individual batches show greater variation as described below.
- **Women's strongest subject is biology** (average rank 1.3); **men's are biology and computer science** (both 1.8). Women's second strongest is computer science (3.0), so women are not weak in computing, but their average there is below men's.

Subject	Men (avg GPA)	Women (avg GPA)	Gap (W – M)	95% interval	Women higher in
Physics	8.16	7.99	-0.18	-0.49 to +0.14	2 of 6 batches
Mathematics	7.47	7.32	-0.15	-0.52 to +0.22	2 of 6 batches
Computer science	8.37	8.25	-0.12	-0.42 to +0.17	1 of 6 batches
Chemistry	8.08	8.16	+0.08	-0.26 to +0.42	3 of 6 batches
Humanities	7.83	7.99	+0.16	+0.06 to +0.26	6 of 6 batches
Earth sciences	7.64	7.88	+0.24	-0.15 to +0.63	4 of 6 batches
Biology	8.26	8.51	+0.25	-0.06 to +0.55	4 of 6 batches

Intervals use the six batch gaps as the sample (a t-interval over batches), so they reflect how much the gap changes from batch to batch, not the number of students.

3. Graphs

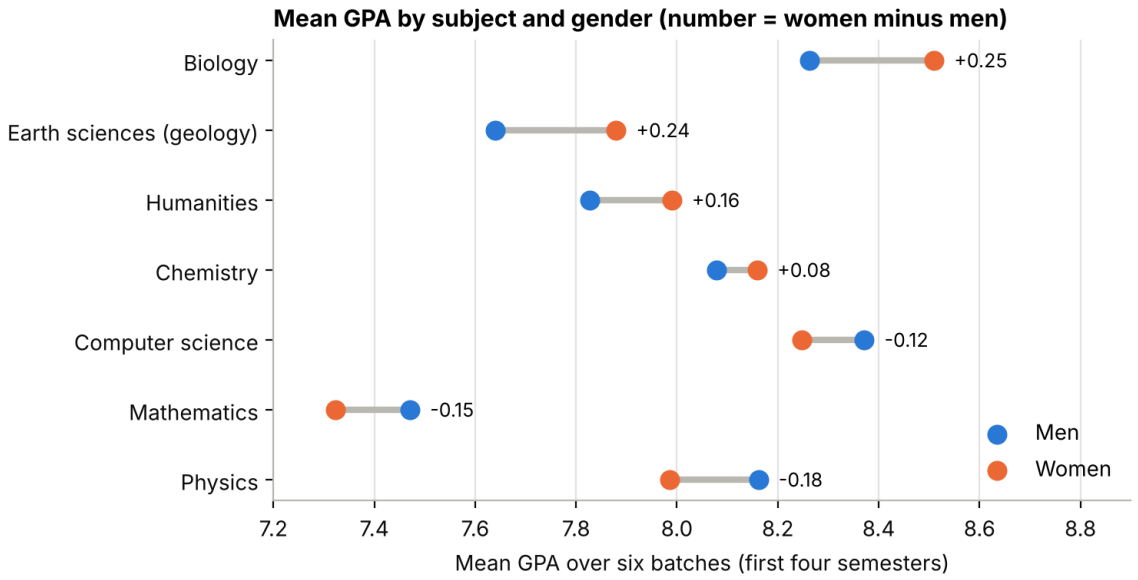


Figure 1. Average GPA of men (blue) and women (orange) in each subject. The number is the gap; for physics, mathematics and computer science it is negative (men higher).

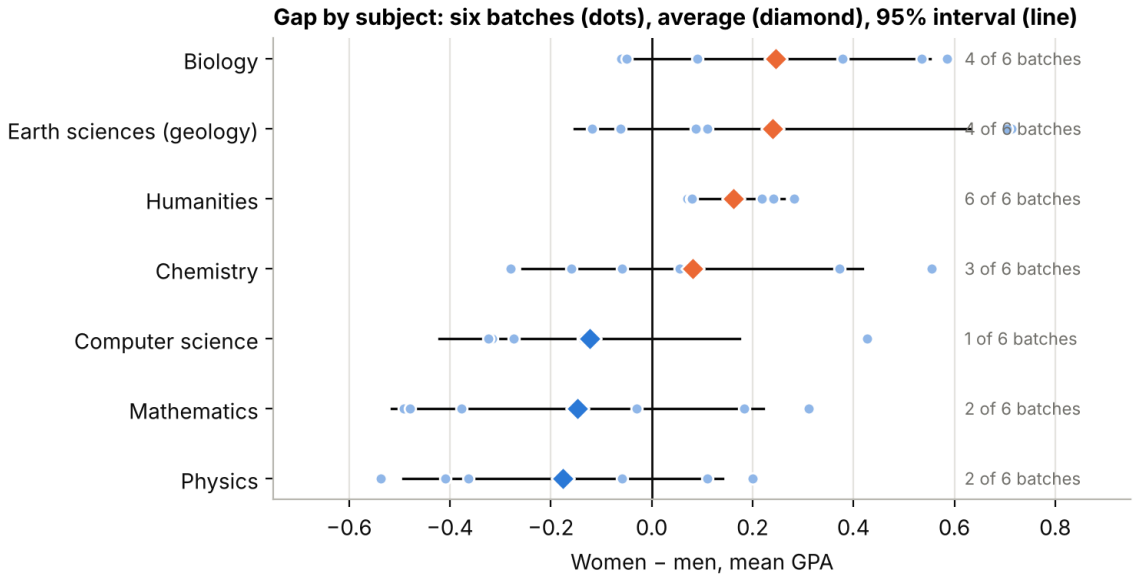


Figure 2. Women − men in each subject. Each light dot is one batch; the diamond is the average; the line is the 95% interval. A line that does not cross zero means a consistent gap (only humanities).

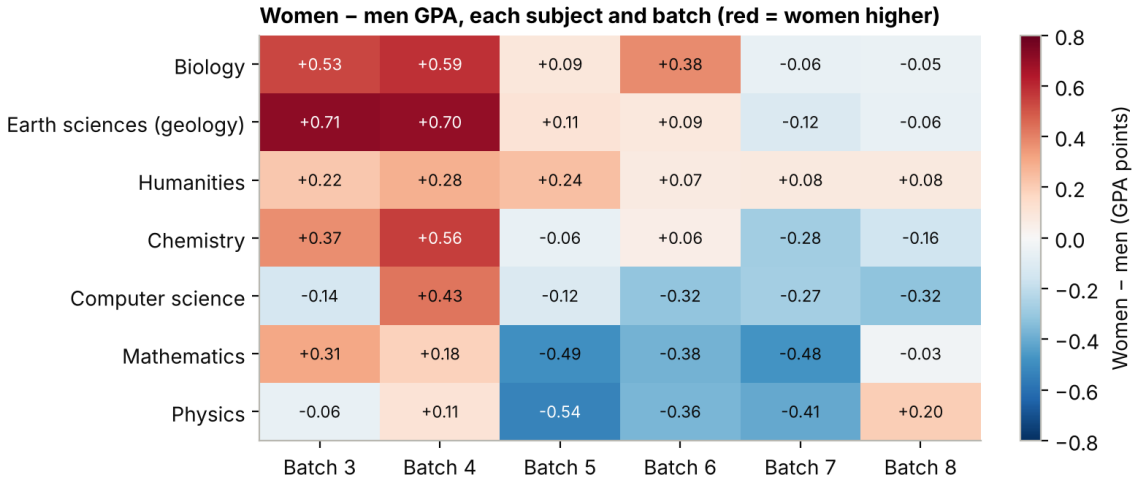


Figure 3. The same gaps for every subject and batch. Red means women higher, blue means men higher. Batches 3 and 4 are mostly red; Batches 5 to 7 are blue in physics, mathematics and computer science.

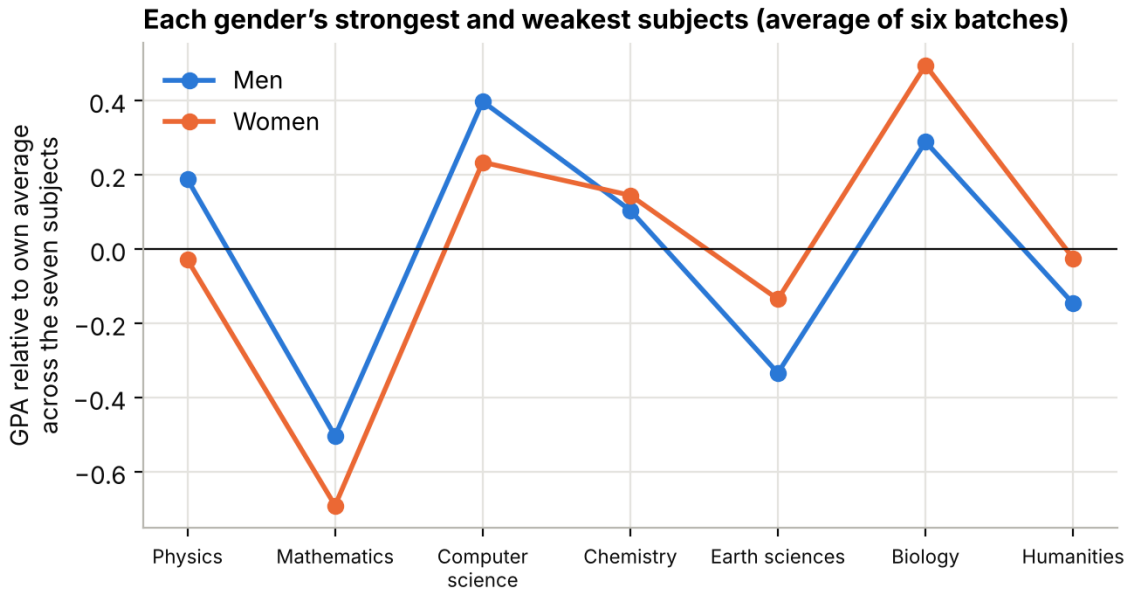


Figure 4. Each gender's GPA in each subject relative to that gender's own average across all seven subjects. A woman's profile peaks in biology and dips in mathematics; the men's profile peaks in computer science and biology and dips in mathematics and earth sciences.

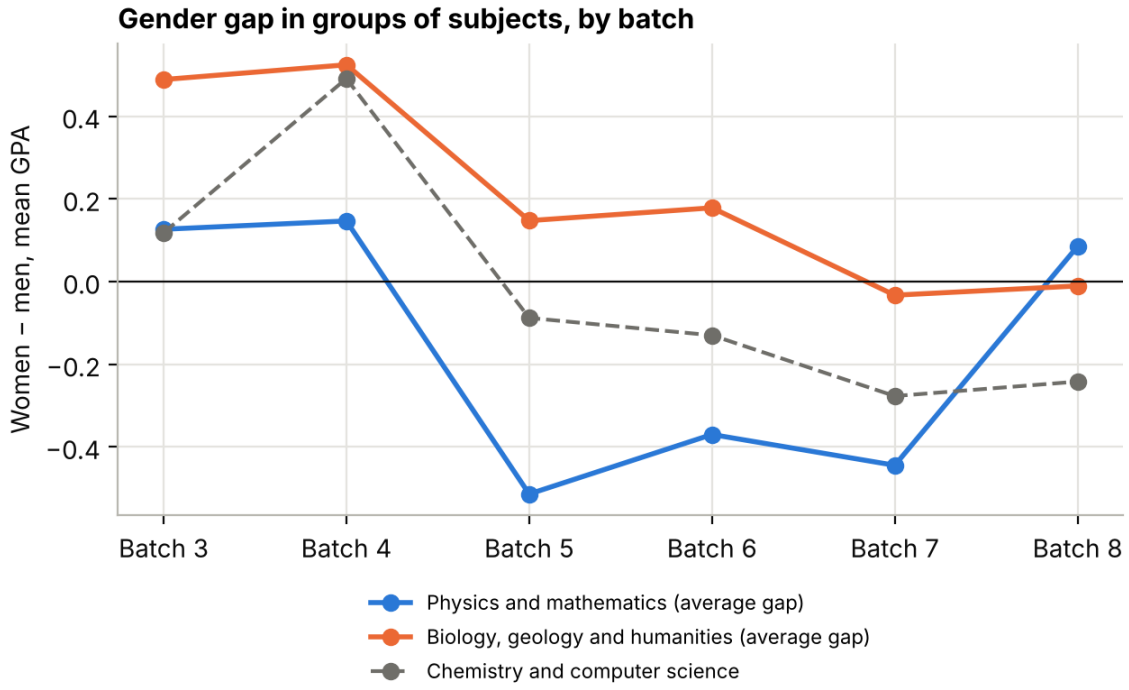


Figure 5. Average gap for three groups of subjects in each batch. The physics-and-mathematics line sits below the biology, geology and humanities line in five of six batches (all but Batch 8).

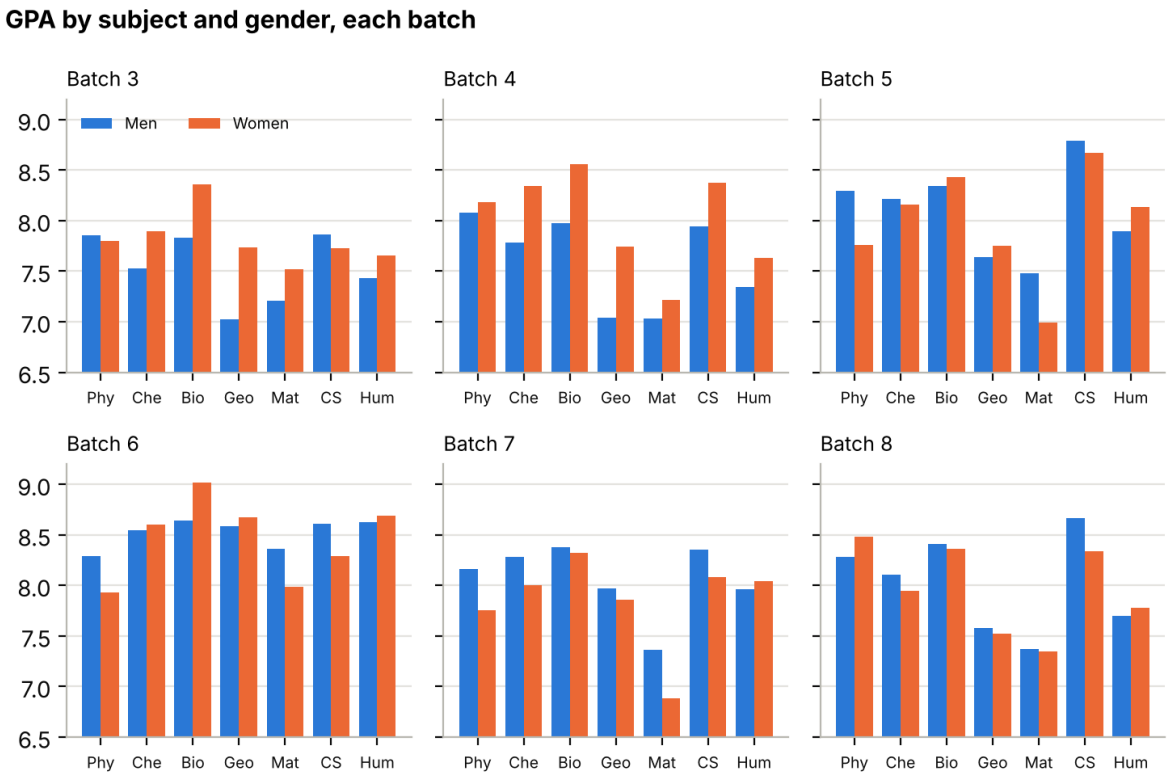


Figure 6. Subject GPA for men and women in each of the six batches. Levels differ widely between batches (Batch 6 is high in every subject), so gaps are more reliable than levels when comparing batches.

4. Could this relate to who chooses which major?

The idea we want to test is that students, consciously or not, pick the major in which they have done well, and that if women's relative strengths lie in biology, earth sciences and humanities while men's lean to physics, mathematics and computer science, the gender mix of majors would follow. Our data are consistent with the first half of that idea and cannot confirm the second:

- **What the data show:** relative to men, women's performance in the first four semesters is weakest in physics, mathematics and computer science and strongest in biology, earth sciences and humanities. This pattern repeats in five of six batches.
- **What the data do not show:** which majors women and men actually chose. Without the number of men and women in each major in each batch, we cannot say whether majors with a negative gap for women have fewer women. The next phase of the study aims to examine that data.
- **Size of the effect:** the gaps are small, about 0.1 to 0.25 GPA points, which is much less than the variation between batches. Differences of this size are unlikely, on their own, to explain a large difference in major choice. Confidence, role models, and expectations about career paths also matter, and they are not in this data.
- **Direction of cause:** a gap in these averages could be a cause of choice (people go where they do well), a result of choice (in semesters 3 and 4 students take subjects they chose, so the later averages already reflect those choices), or neither. Section 5 explains why.

5. Limits

- **Averages only.** We have the mean GPA for each gender, subject and batch, with no counts or spread. A mean over 30 women is much less stable than a mean over 100 men, and we cannot account for that. Given that the percentage of female UG students at IISER Kolkata is already around 27%, the data is likely to show more fluctuations.
- **Six batches, so each result is based on six numbers.** That is the reason that only humanities reaches a conventional level of significance, and that this single result is one of seven subjects tested. With seven subjects, one result with p below 0.05 would be expected about a third of the time by chance alone.
- **Selection by choice.** In semesters 3 and 4 students take the three subjects they chose. A student who is weak in mathematics is less likely to take it again, so GPA in mathematics after semester 2 is measured on a more selected group of students. This affects men and women differently if they choose differently.
- **Subjects are graded differently.** A GPA of 7.3 in mathematics and one of 8.3 in biology are not the same level of performance if the courses are graded differently. We therefore compare women with men within each subject, not subjects with each other.
- **Not student-level.** We cannot adjust for entrance rank, school background or reservation category, and we do not know whether the same students are in each subject.
- **No link to later years.** These are the first four semesters only; our CGPA report covers all ten semesters.

6. Next Phase of the Study

- **Comparison with the number of men and women in each batch**, and in each subject in semesters 3 and 4 (so we can judge uncertainty).
- **The number of men and women choosing each major in each batch.** This is the one table that lets us test whether the subject gaps relate to major choice.
- **Checking the correlation** between the department's male-to-female ratio and the department's male-to-female GPA ratios (as recommended by Dr. Rangeet Bhattacharyya)
- **Subject GPA separately for semesters 1 and 2**, when everyone takes every subject. This removes the selection problem and gives the cleanest comparison.
- **The standard deviation, or quartiles, of GPA** for each gender, subject and batch.